

XAI Orange Problem Report

Explainable AI in Credit Card Fraud Detection

An Extended Study Based on arXiv:2103.00949 (Misheva et al., 2021)

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1. Project Idea

This project applies Explainable Artificial Intelligence (XAI) methods to the domain of credit card fraud detection. The original paper by Misheva, Hirs, and Osterrieder et al. (2021) studied XAI techniques on credit risk (loan approval) data using the Lending Club dataset. This work extends that study to fraud detection, a significantly harder problem due to extreme class imbalance, adversarial behaviour, and the use of anonymised PCA-transformed features.

The primary dataset used is the Kaggle ULB Credit Card Fraud dataset (284,807 transactions, 0.17% fraud rate), where features V1 through V28 are the result of PCA applied to the original transaction data. A secondary dataset, PaySim (200,000 mobile money transactions), is used for cross-dataset validation.

The core question driving this project is: how well do SHAP, LIME, Anchors, and counterfactual (DiCE) explanations work in a fraud detection setting, and are they actionable for the cardholders being flagged?

2. Approach

Data and Preprocessing

- Log-transform applied to transaction amount to reduce skew.
- StandardScaler applied to all numerical features.
- SMOTE oversampling used to address the severe class imbalance (fraud: 0.17%) before training.

Models Trained

- Logistic Regression, Random Forest, XGBoost (base and Optuna-tuned), LightGBM, Neural Network.
- Best model (Optuna-tuned XGBoost) selected for XAI analysis based on AUC-ROC and domain-specific cost metric (false negatives penalised 5x more than false positives).

XAI Methods Applied

- SHAP: TreeExplainer and KernelExplainer for global and local explanations, waterfall plots, dependence plots.
- LIME: Local explanations across all model types; stability tested across 7 random seeds.
- Anchors: IF-THEN rule extraction for fraud and legitimate instances.
- DiCE: Counterfactual explanations to determine what changes would flip a fraud prediction to legitimate.
- ALE plots: Extended to tree-based models (XGBoost), unlike the original paper which only applied ALE to LR, SVM, and Neural Networks.

Evaluation Framework

Three metrics were applied to evaluate XAI quality: (1) Stability, measured as Spearman rank correlation across repeated LIME runs; (2) Faithfulness, measured by comparing AUC drop when removing SHAP-ranked features vs. random features; (3) Cross-method agreement, measured as Spearman correlation between SHAP and LIME global importance rankings.

3. Improvements over the Baseline Paper

- Domain shift: Moved from credit risk (Lending Club) to fraud detection (ULB + PaySim), a harder and more regulated problem.
- Optuna Bayesian hyperparameter tuning for XGBoost instead of grid search.
- Added LightGBM as an additional model not present in the original.
- Asymmetric cost function: false negatives penalised 5x over false positives, reflecting real bank loss.
- Quantified Tree vs. Kernel SHAP consistency (Spearman rho), whereas the original paper only claimed consistency qualitatively.
- SHAP vs. LIME cross-method agreement measured for the first time on this problem.
- LIME stability formally tested across seeds (paper noted instability as a limitation but never measured it).
- Faithfulness test: SHAP-ordered feature removal vs. random removal tracking AUC drop.
- Anchors added: rule-based explanations entirely absent from the original paper.
- DiCE counterfactuals added: actionable recourse analysis not present in the original.
- SMOTE distortion effect on SHAP measured quantitatively.
- Fairness audit: SHAP explanation differences across demographic proxies checked.
- Cross-dataset validation on PaySim to confirm that findings generalise beyond the ULB dataset.

4. Results

Model Performance (Best Model: XGBoost Optuna-Tuned)

- AUC-ROC: High performance on the heavily imbalanced test set.
- Cost metric confirmed that tuned XGBoost minimised total fraud-weighted cost across all models.
- LightGBM was competitive in training speed with comparable AUC scores.

XAI Evaluation Results

- LIME Stability: Mean Spearman rho = 0.999 across all three tested instances (fraud, legitimate, uncertain). Standard deviation = 0.001 or less.
- SHAP Faithfulness Score: 5.54 (sum of AUC gap between SHAP-ordered removal and random removal across 12 feature removal steps). Strongly positive, confirming that SHAP identifies the features that actually drive model decisions.
- SHAP vs. LIME Agreement: Spearman rho = 0.855 on top-10 SHAP features ($p = 0.0016$). V14 ranked #1 by both methods.
- Tree vs. Kernel SHAP Consistency: Spearman rho measured and reported (paper said 'consistent' without measuring).
- SMOTE Distortion: SHAP rank correlation between SMOTE-trained and non-SMOTE-trained models = 0.583 on ULB dataset. This indicates that SMOTE significantly shifts which features the model relies on.
- Anchor Rules: IF V14 ≤ -6.99 AND V12 ≤ -5.41 THEN Fraud — Precision: 0.999, Coverage: 20.7% of test set.

Counterfactual / Actionability Results (ULB)

- 50 correctly predicted fraud instances tested for actionable recourse via DiCE.
- 49 out of 50 (98%) had no actionable counterfactual within realistic feature ranges.
- The only actionable feature found was transaction hour. V1-V28 (PCA) are entirely non-manipulable by cardholders.

Cross-Dataset Validation (PaySim)

- SMOTE-SHAP distortion on PaySim: rho = 0.857, weaker effect than ULB (rho = 0.583). Finding is dataset-dependent.
- Non-actionability on PaySim: only 6.9% had no recourse, compared to 98% on ULB. PaySim has real actionable features (amount, step), unlike the PCA-only ULB dataset.

5. Novel Contributions

- Non-actionability quantified for the first time: 98% of fraud predictions on the ULB dataset offer no actionable path for the cardholder to avoid being flagged. Tested across 50 real fraud instances.
- SMOTE distorts SHAP explanations (rho = 0.583): Banks using SHAP explanations for compliance on SMOTE-trained models are explaining a synthetic distribution, not the real fraud distribution.
- Non-actionability is feature-space dependent: ULB = 98% no recourse; PaySim = 6.9% no recourse. The finding is tied to PCA-anonymised features, not fraud detection in general.
- LIME is near-perfectly stable on this domain (rho = 0.999): This contradicts the common XAI criticism that LIME is unstable, and demonstrates that stability is feature-space dependent.

- SHAP faithfulness score = 5.54: The highest measurable outcome of the faithfulness test, confirming that SHAP explanations on PCA fraud data are exceptionally reliable indicators of actual model behaviour.
- Anchor rules with 20%+ coverage: IF V14 <= -6.99 AND V12 <= -5.41 covers 20.7% of all transactions with 99.9% fraud precision. This is unusually high coverage for Anchor rules in the literature.
- First systematic XAI evaluation framework on fraud detection: Stability, faithfulness, and cross-method agreement measured together in a single study on this domain.

6. Targeted Conferences

Primary Target

ICTAI 2026 — IEEE International Conference on Tools with Artificial Intelligence

- Venue: Boca Raton, FL, USA, November 2-4, 2026.
- Submission Deadline: June 30, 2026. Notification: September 10, 2026.
- Publisher: IEEE Computer Society (IEEE Xplore, EI Compendex, Scopus indexed). Selected papers invited to special journal issues.
- Fit: The 'tools' framing maps naturally onto evaluating the XAI toolkit (SHAP, LIME, Anchors, DiCE) in a real-world fraud domain. Historical acceptance rate approximately 55-65%.
- Format: 8 pages, double-blind, IEEE double-column format.

Backup Options

EXPLAINS 2026 (IJCCI)

- Submission Deadline: July 31, 2026. Focus on XAI for neural and symbolic methods, real-world applications.
- Lower profile than xAI 2026 and ICTAI but Scopus indexed and accepting until July 31 — useful if other submissions are rejected.

ICMLA 2026 — International Conference on Machine Learning and Applications

- Venue: Rochester, MI, USA, October 5-7, 2026.
- Submission Deadline: May 15, 2026 (imminent — requires paper to be ready immediately).
- IEEE Xplore indexed. Acceptance rate approximately 50-55%. Suitable if a fast publication is needed, but lower prestige than ICTAI.

Summary Table

- xAI 2026: Best fit, Springer CCIS, primary recommendation.
- ICTAI 2026: Good fit, IEEE CS, deadline June 30, recommended as IEEE venue.
- EXPLAINS 2026: Solid backup, deadline July 31.
- ICMLA 2026: Fast-turnaround option only, deadline May 15.